ALAMEIN INTERNATIONAL UNIVERSITY
Faculty of Computer Science & Engineering
Knowledge Based Systems | AIE212

Medical Diagnosis Expert System

Clinical Triage & Preliminary Diagnosis KBS

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PHASE 1: ANALYZE THE PROBLEM

1.1 Problem Context & Real-World Setting

One of the biggest problems in healthcare today is that the number of patients is growing faster than the number of doctors available to see them. This affects clinics, emergency departments, and telemedicine services alike. Clinical triage — basically deciding which patients need to be seen first based on how serious their condition is — often becomes a bottleneck. When triage is slow or inconsistent, patients can deteriorate while waiting, which in serious cases can be fatal.

Our system is meant to work as a knowledge-based triage and early diagnosis assistant. We designed it to be useful in several different healthcare settings:

- **Primary Healthcare Clinics:** GP clinics, where a nurse or receptionist does a quick intake before the doctor is available — the system can help flag which patients need to be seen urgently.
- **Telemedicine Platforms:** Telemedicine platforms, where patients describe their symptoms online or through an app. The system can help decide whether the patient can be managed remotely or needs to come in physically.
- **Emergency Department Triage:** Emergency departments, which often deal with a lot of patients at once. The system gives triage nurses a second opinion by matching symptoms and vitals to known conditions.
- **Rural & Underserved Health Posts:** Rural or underserved health posts, where there may be no doctor on site and a paramedic or community health worker is doing the assessment. The system provides structured decision support without needing a specialist present.

The system is not meant to replace doctors — that's not realistic or safe. The goal is to give healthcare workers a reliable tool to support their early decision-making, help avoid cases where urgency is either over- or under-estimated, and make sure patients get to the right level of care without unnecessary delays.

1.2 Patient Inputs

The system takes structured inputs from the patient (or the person assessing them). We grouped these into four categories, which roughly follow how a clinical consultation actually works:

Input Category	Specific Input Parameters
Chief Complaint & Symptoms	Primary complaint (free-select from symptom list); duration of symptoms (hours / days / weeks); onset type (sudden vs gradual); severity rating (1–10 self-reported scale); symptom progression (improving / stable / worsening)
Vital Signs & Physical Indicators	Body temperature (°C); heart rate (bpm); blood pressure (systolic / diastolic mmHg); respiratory rate (breaths/min); oxygen

	saturation SpO2 (%); level of consciousness (alert / confused / unresponsive)
Patient Background Information	Age; biological sex; body weight / BMI; known chronic conditions (diabetes, hypertension, asthma, cardiac disease, etc.); current medications; known drug allergies; smoking / alcohol status
Symptom-Specific Detail	Location of pain (if applicable); type of pain (sharp / dull / burning / pressure); associated symptoms (nausea, vomiting, dizziness, shortness of breath, chest tightness, rash, swelling, bleeding); recent travel history; recent sick contacts or exposure
Recent Medical History	Recent hospitalizations; recent surgeries or procedures; recent laboratory or imaging results (if available); vaccination status; pregnancy status (if applicable)

1.3 Expected Outputs

At the end of each assessment, the system gives the following outputs:

- **Primary Differential Diagnosis:** The most likely condition based on what was entered (e.g., 'Suspected Acute Myocardial Infarction' or 'Likely Tension Headache').
- **Ranked Differential List:** Up to three other possible diagnoses, each with a confidence level (High / Medium / Low) since real cases often aren't clear-cut.
- **Urgency / Triage Level:** A triage level using 4 categories: Immediate (needs action in minutes), Urgent (within 1–2 hours), Semi-Urgent (within 24 hours), or Non-Urgent (can wait for a routine appointment).
- **Recommended Action:** A clear recommendation of what to do, like 'Call emergency services immediately', 'Refer to a cardiologist', 'Book a GP appointment', 'Rest and increase fluids', etc.
- **Red Flag Alerts:** Warnings that get triggered when inputs suggest something serious — for example, SpO2 under 90%, chest pain radiating to the arm, or the patient being unresponsive. These automatically push the urgency to Immediate regardless of anything else.
- **Reasoning Explanation:** A readable explanation of why the system reached its conclusion — which symptoms, vitals, and background factors played a role. This is important so a clinician can actually review the logic and not just trust a black box.

1.4 Realistic Diagnostic Scenarios

Below are some example cases ranging from minor to life-threatening. These represent the kinds of scenarios our system needs to handle correctly:

Scenario Type	Description & Expected System Response
NON-URGENT — Common Cold /	Patient: 28-year-old, reports runny nose, mild sore throat, sneezing for 3 days, temperature 37.4°C, normal vital signs, no chronic conditions. System output: Probable URTI / common cold. Urgency: Non-Urgent. Action: Advise

Upper Respiratory Tract Infection	rest, fluids, OTC antihistamines/decongestants. No referral needed. Confidence: High.
NON-URGENT — Tension Headache	Patient: 35-year-old office worker, bilateral pressing headache, severity 4/10, duration 5 hours, worsened by screen use, no fever, no neurological symptoms, no neck stiffness. System output: Tension-type headache. Urgency: Non-Urgent. Action: OTC analgesia (paracetamol/ibuprofen), posture advice, GP follow-up if recurrent.
SEMI-URGENT — Urinary Tract Infection (UTI)	Patient: 24-year-old female, burning urination, frequent urge to urinate, mild lower abdominal pain for 2 days, temperature 37.8°C, no flank pain, no fever above 38°C, no known renal disease. System output: Probable UTI (lower). Urgency: Semi-Urgent. Action: Urine dipstick test; initiate empirical antibiotic therapy; GP visit within 24 hours.
URGENT — Suspected Appendicitis	Patient: 19-year-old male, progressive right lower quadrant pain for 12 hours, pain severity 7/10, nausea, low-grade fever 38.2°C, pain worsens on movement. System output: Suspected acute appendicitis. Urgency: Urgent. Action: Refer to emergency department immediately for surgical evaluation; nil by mouth; imaging required.
IMMEDIATE — Acute Myocardial Infarction (Heart Attack)	Patient: 58-year-old male, known hypertension and diabetes, sudden crushing chest pain radiating to left arm, severity 9/10, sweating, shortness of breath, SpO2 94%, heart rate 102 bpm. System output: RED FLAG — Suspected Acute MI. Urgency: Immediate. Action: Call emergency services (ambulance); aspirin 300mg if not contraindicated; do not leave patient unattended. Confidence: Very High.
IMMEDIATE — Stroke (FAST Protocol Positive)	Patient: 67-year-old female, sudden facial drooping on right side, arm weakness, slurred speech, onset 45 minutes ago. System output: RED FLAG — Suspected Ischemic Stroke. Urgency: Immediate. Action: Emergency services immediately; time-critical — thrombolysis window active; do not give food or water.

1.5 System Scope & Limitations

In Scope

- **Target conditions:** Around 40 conditions covering cardiovascular, respiratory, neurological, GI, urological, and general medicine. We focused on both common presentations and serious/high-risk ones.
- **Patient age range:** Adults aged 18–80. We didn’t include pediatric or neonatal cases since those require completely different diagnostic criteria.
- Triage and preliminary diagnosis based on symptoms and vital signs entered through a structured form.
- Red flag detection for immediate life-threatening presentations.
- Four-level urgency classification (Immediate / Urgent / Semi-Urgent / Non-Urgent), consistent with standard triage protocols.
- Differential diagnosis with ranked confidence levels.
- Transparent reasoning explanation for each output.

Out of Scope / Limitations

- **No definitive diagnosis:** It only gives preliminary differentials and a triage level — it doesn't confirm diagnoses. A licensed clinician must review and validate everything the system outputs.
- **No physical examination:** The system can't do physical exams. Things like listening to the chest or palpating the abdomen need a real clinician — any findings from that have to be entered manually.
- **No laboratory or imaging interpretation:** It doesn't interpret blood results, ECGs, X-rays, or any other investigations. Those need to be analyzed by a professional.
- **No pediatric coverage:** Patients under 18 are not supported. Pediatric diagnoses work differently and we haven't encoded those rules in this version.
- **No prescribing authority:** The system suggests drug categories and general management steps but doesn't write prescriptions or calculate specific doses.
- **No continuous monitoring:** Each assessment is a snapshot. The system doesn't track changes over time or connect to live monitoring devices.
- **Rare and complex conditions:** The knowledge base only covers common and high-acuity cases. Rare conditions, unusual presentations, or patients with multiple interacting problems may not get an accurate result.
- **Medico-legal disclaimer:** This is a decision-support tool only. Full clinical responsibility stays with the doctor or healthcare professional on site.